

WEEK 4 ASSIGNMENT

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

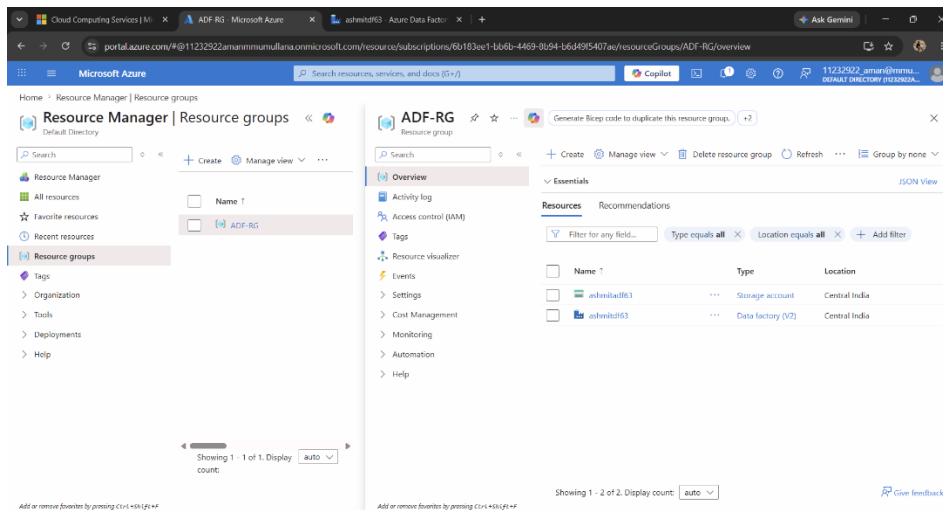
Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks

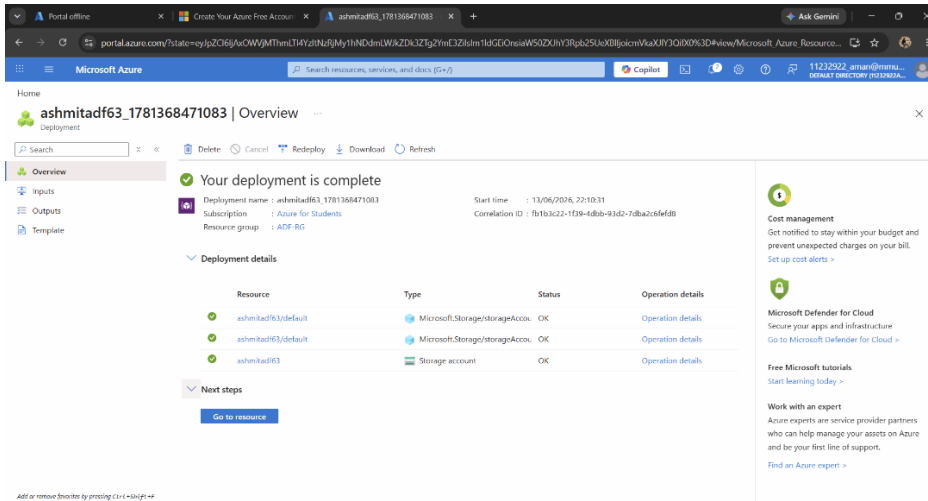
Task 1:

- Explored Azure Portal
- Created a Resource Group

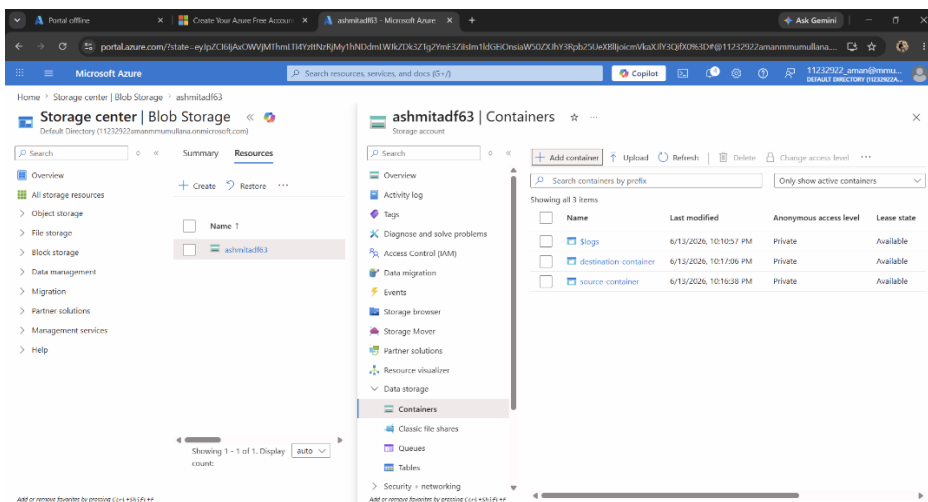


Task 2: Storage Setup

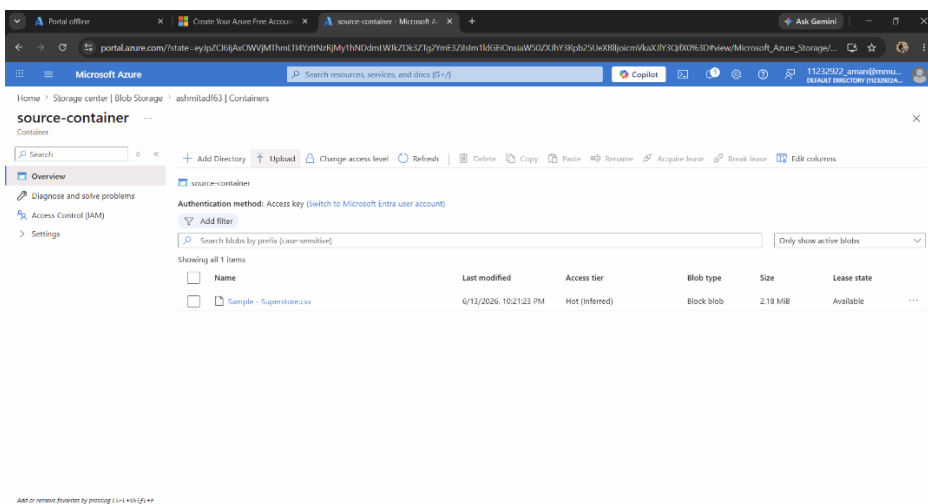
- Create a Storage Account



- Create a Blob Container

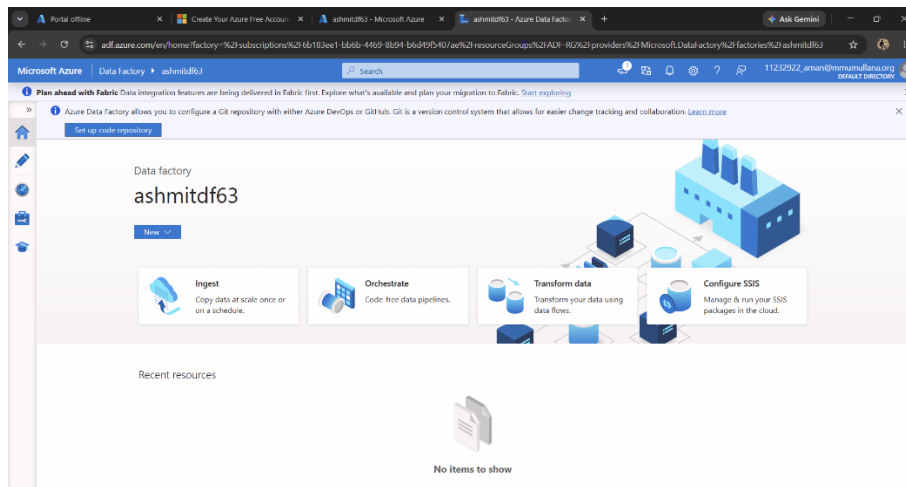


- Upload a CSV file

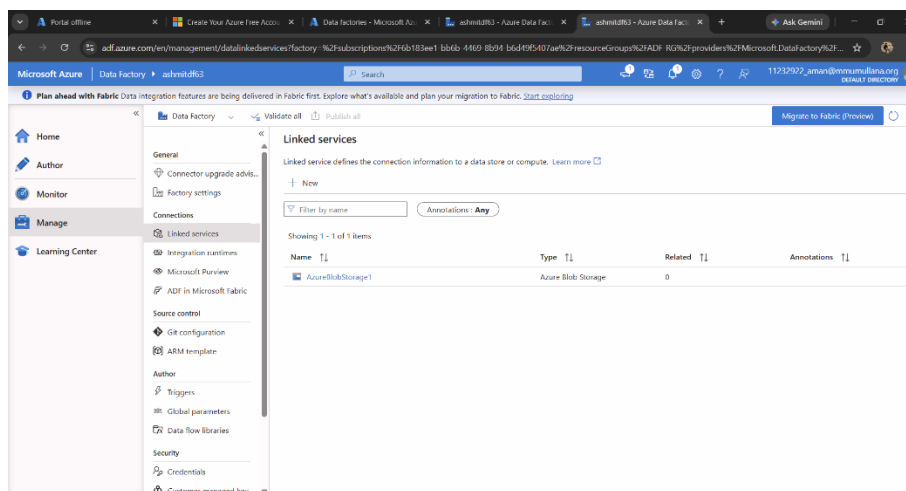


Task 3: ADF Basics

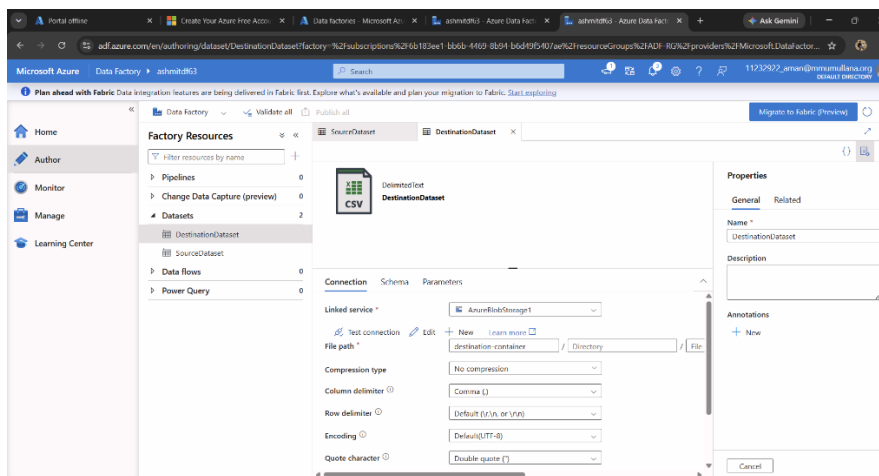
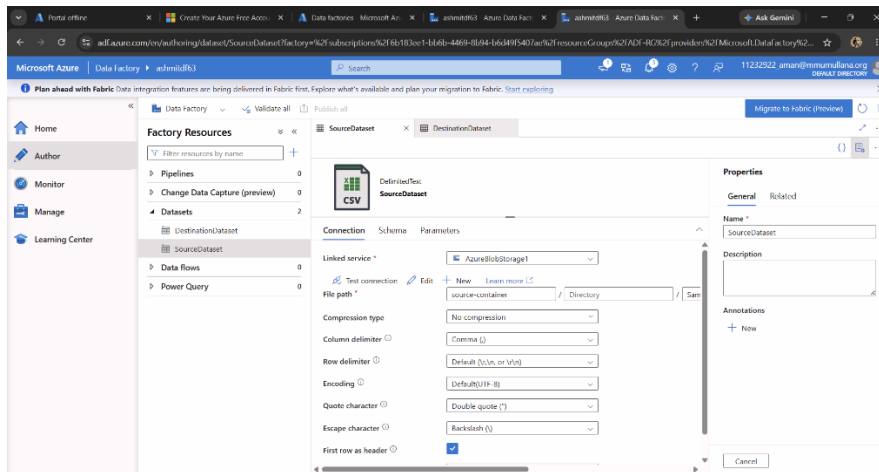
- Create Azure Data Factory



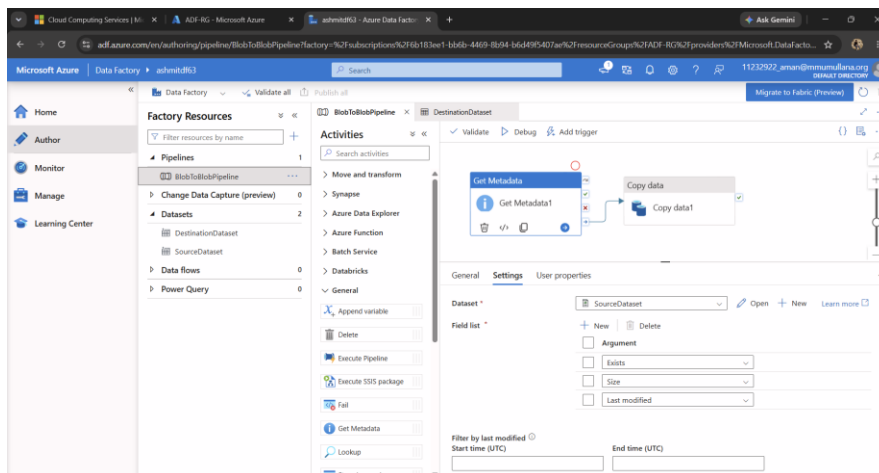
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)



- Create datasets (source + destination)

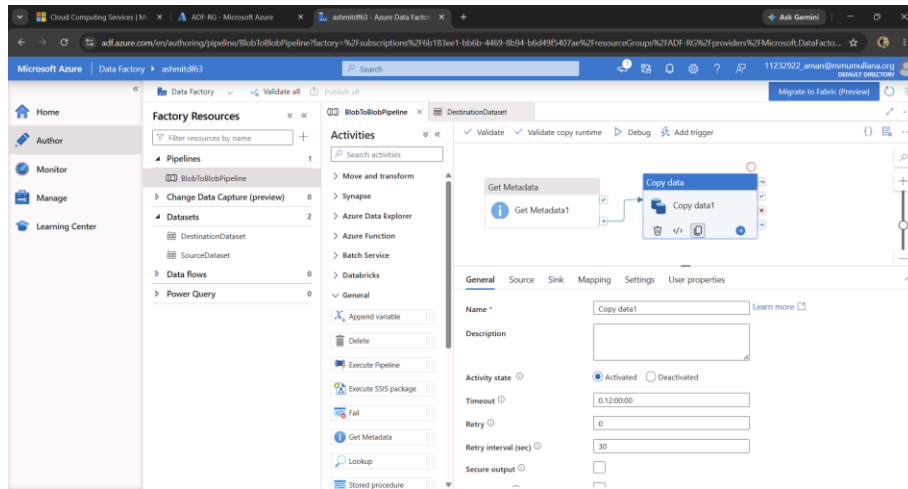


- Use Get Metadata activity

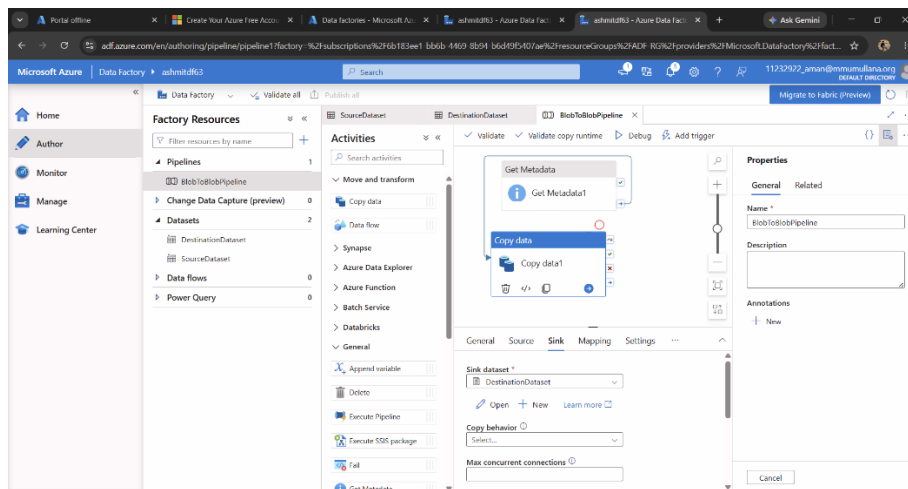
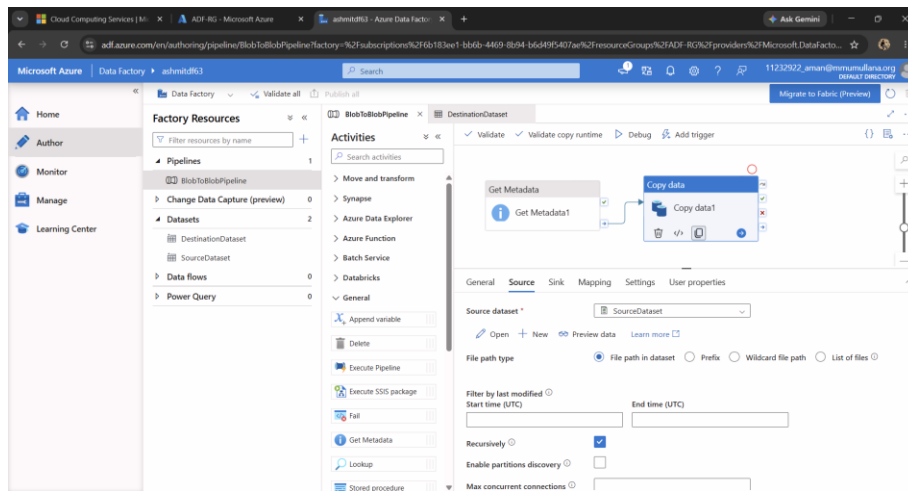


Task 4: Pipeline Development

- Create pipeline using Copy Data activity

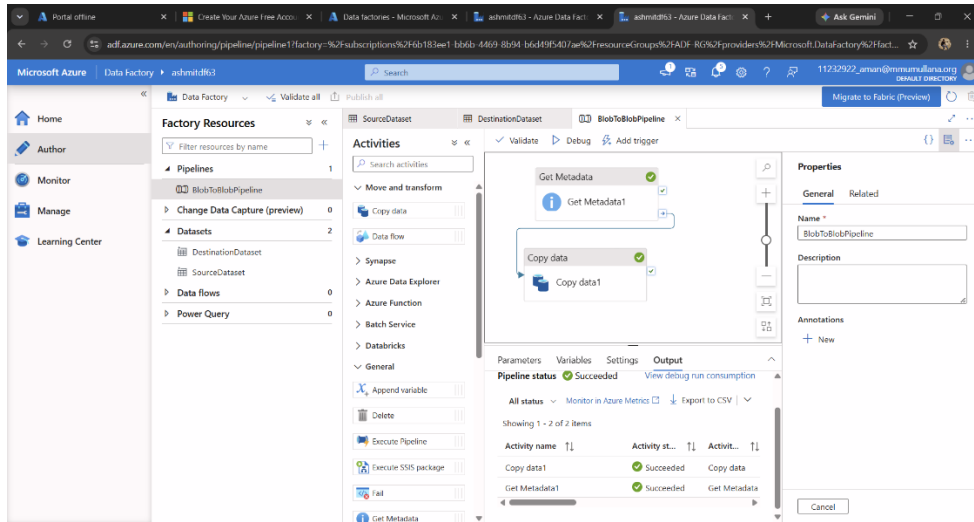


- Configure source and destination



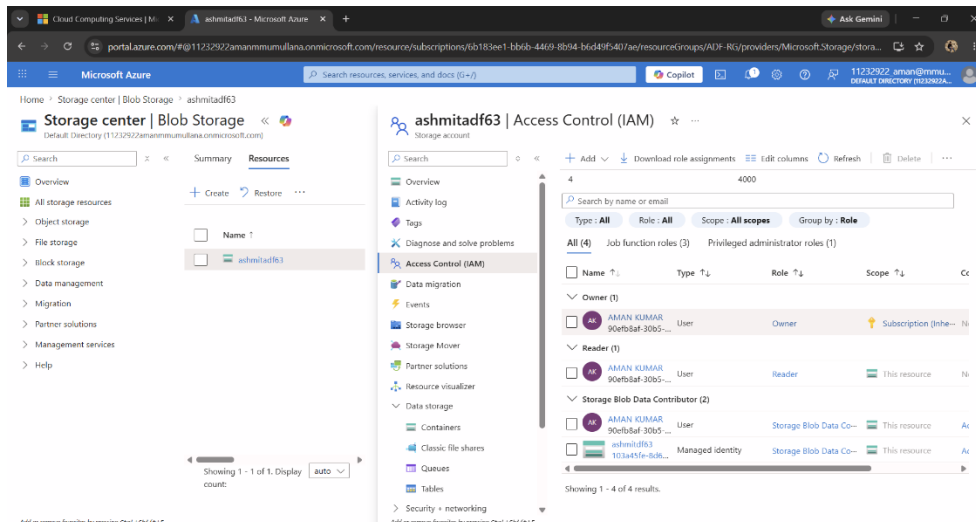
Task 5: Pipeline Execution

- Run pipeline using
Debug/Trigger



Task 6: IAM Roles

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage



Mini Project (Final Task)

Problem Statement:

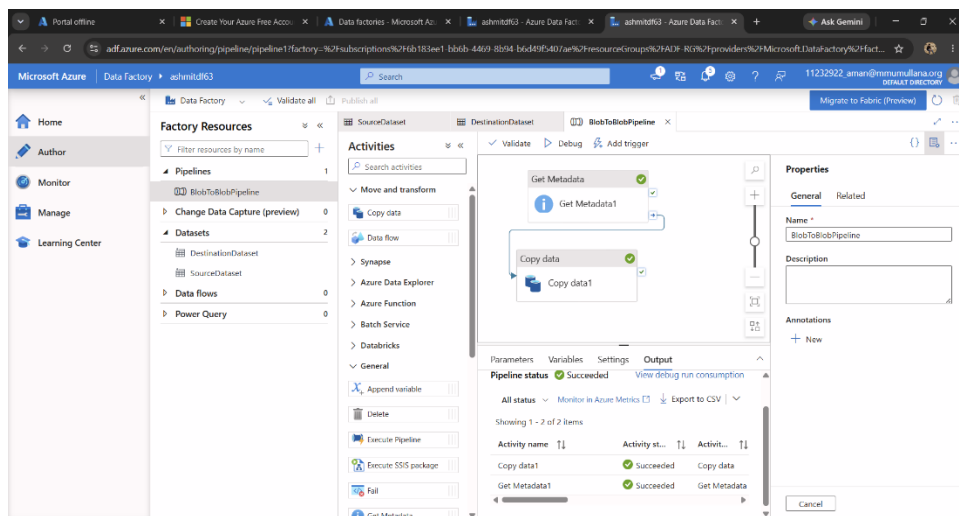
Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

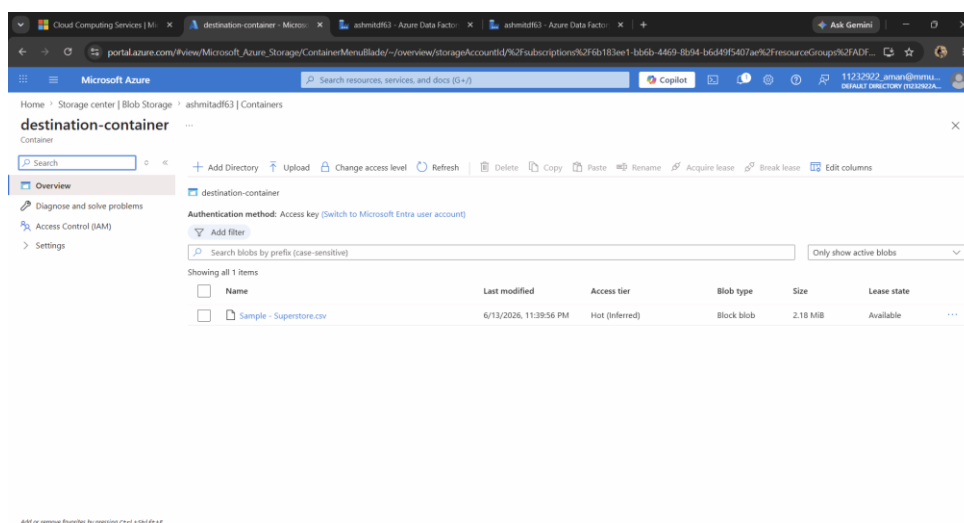
- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

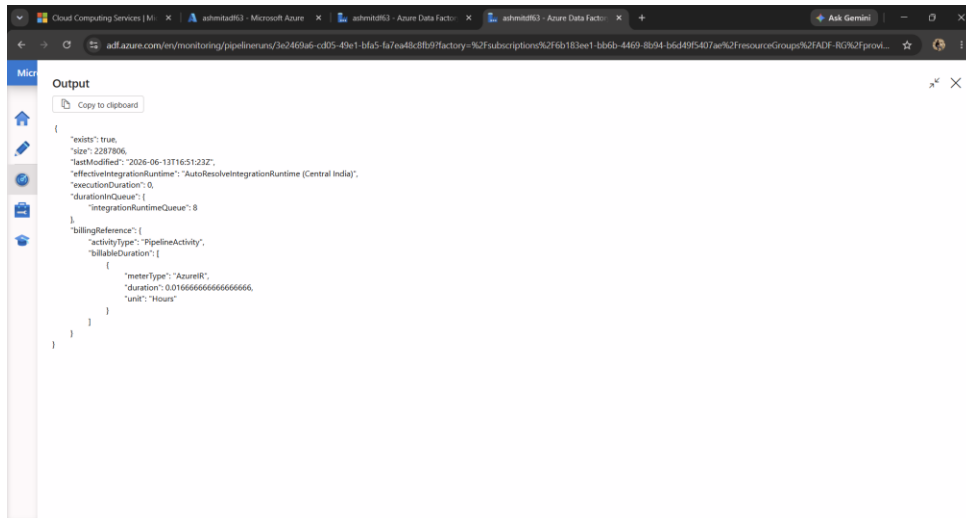
- Pipeline executed successfully



- Data copied to destination



- Metadata validated



Summary of the Assignment

This project demonstrates the creation of an end-to-end data pipeline using Azure Data Factory and Azure Blob Storage. A Storage Account and Blob containers were created, and a CSV file was uploaded to the source container. Azure Data Factory was configured with the required Linked Services and datasets. A pipeline was developed using the **Get Metadata** activity to validate the source file and the **Copy Data** activity to transfer it to the destination container. The pipeline was executed successfully, and its performance was monitored through ADF. IAM roles were assigned to ensure proper access between Azure Data Factory and Azure Storage. This project provided hands-on experience with Azure cloud services, data integration, and workflow management.